

Online Experiments for Language Scientists

Lecture 9: Iterated learning

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Drop-in labs for help with final assessment (coding project, due 4th December)

- Times on <https://jennifer-culbertson.github.io/oels2025/>
- We can help with basic coding stuff
- No questions after 10am on Monday 2nd

Beckner et al (2017)



Clay Beckner
(now at Warwick)

Beckner, C., Pierrehumbert, J., & Hay, J. (2017). The emergence of linguistic structure in an online iterated learning task. *Journal of Language Evolution*, 2, 160–176.

- Design:
 - Iterated artificial language learning experiment
- Question:
 - Does compositional structure emerge ‘for free’ from person-to-person transmission?



Janet Pierrehumbert
(Oxford)



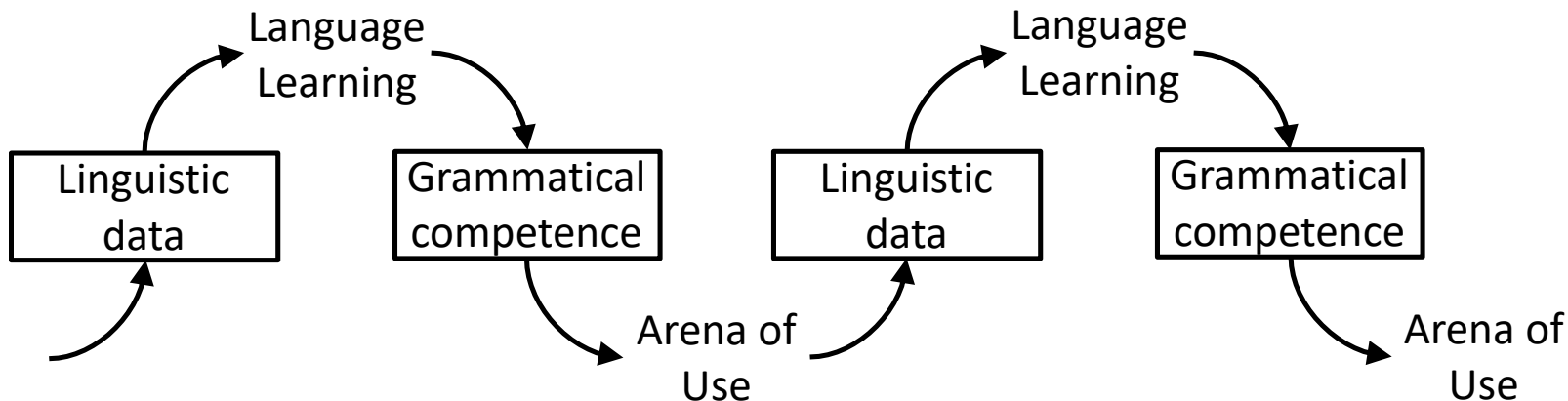
Jen Hay
(Canterbury, NZ)

Structure from transmission?

Language is transmitted via repeated **learning** and **use**

Language is shaped by these processes

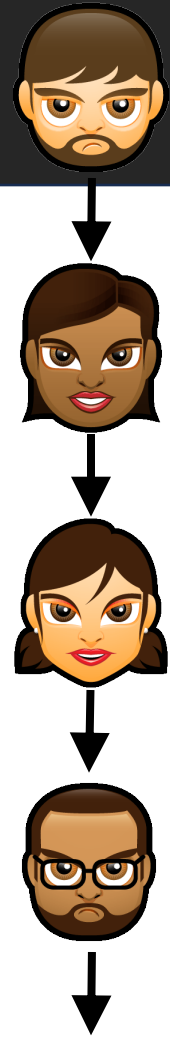
The cycle of learning and use produces structure

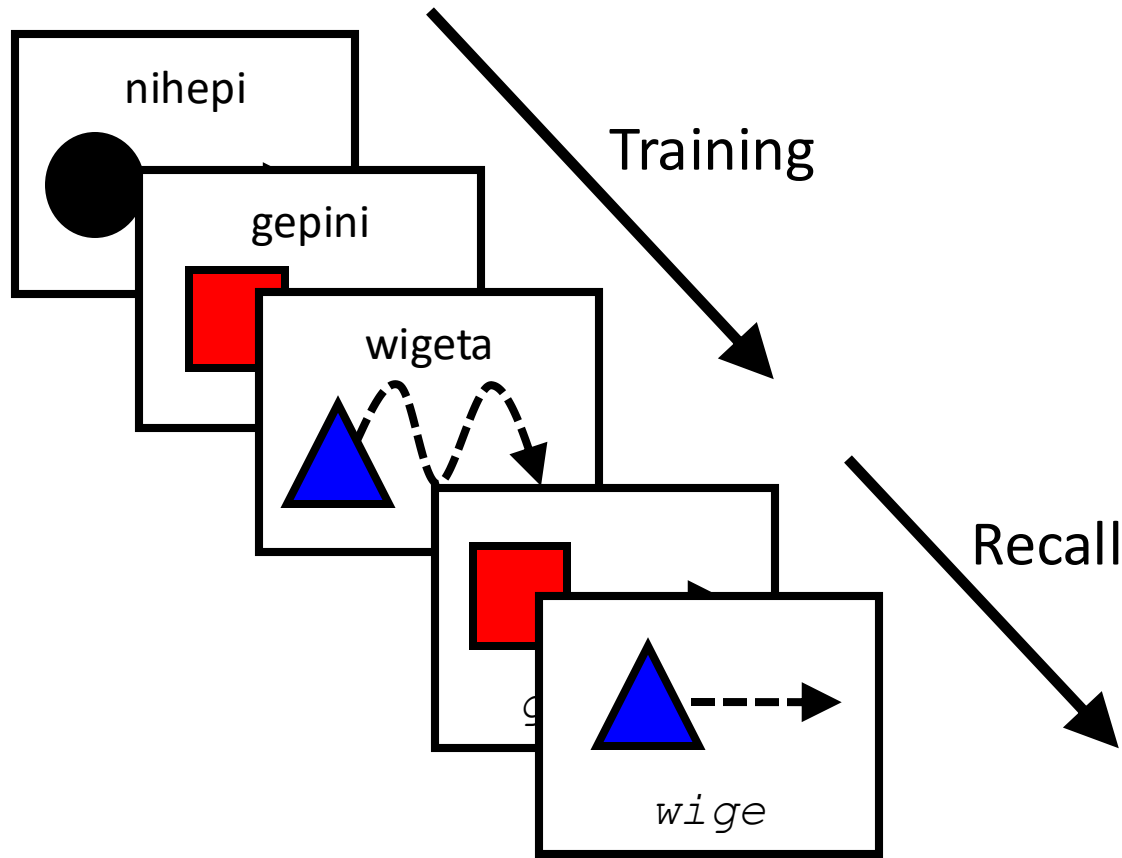


Iterated learning

Transmission over generations can be *simulated* in the lab

- Generations:
 - Individual learners, trained on the output of previous learners
- Chains:
 - A sequence of generations (aka participants)





Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *PNAS*, *105*, 10681-10686.

Demo using this week's practical code

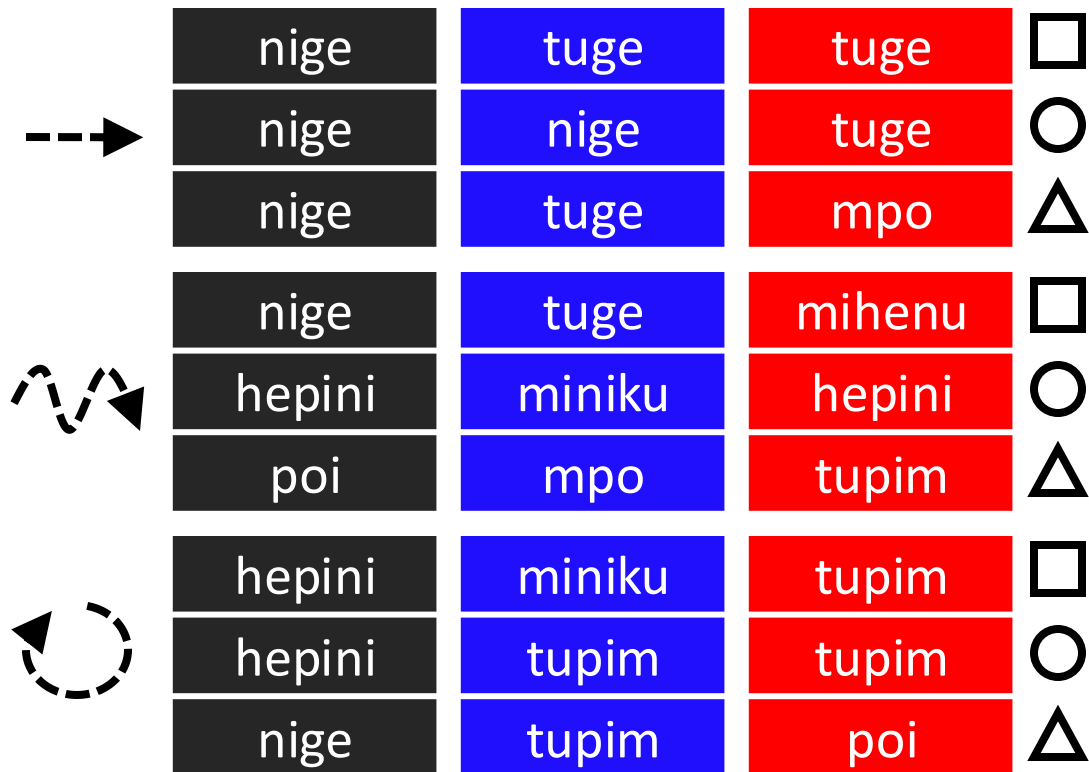
Initial holistic language (chain 4)

-->	wimaku	miniki	gepinini	□
	nihepi	wigemi	mahekuki	○
	wikima	nipikuge	hema	△
↗↘	miwiniku	pinipi	kihemiwi	□
	kinimapi	wikuki	kikumi	○
	miwimi	nipi	wige	△
↻	gepihemi	kunige	miki	□
	pikuhemi	kimaki	pimikihe	○
	mihe	winige	kinimage	△

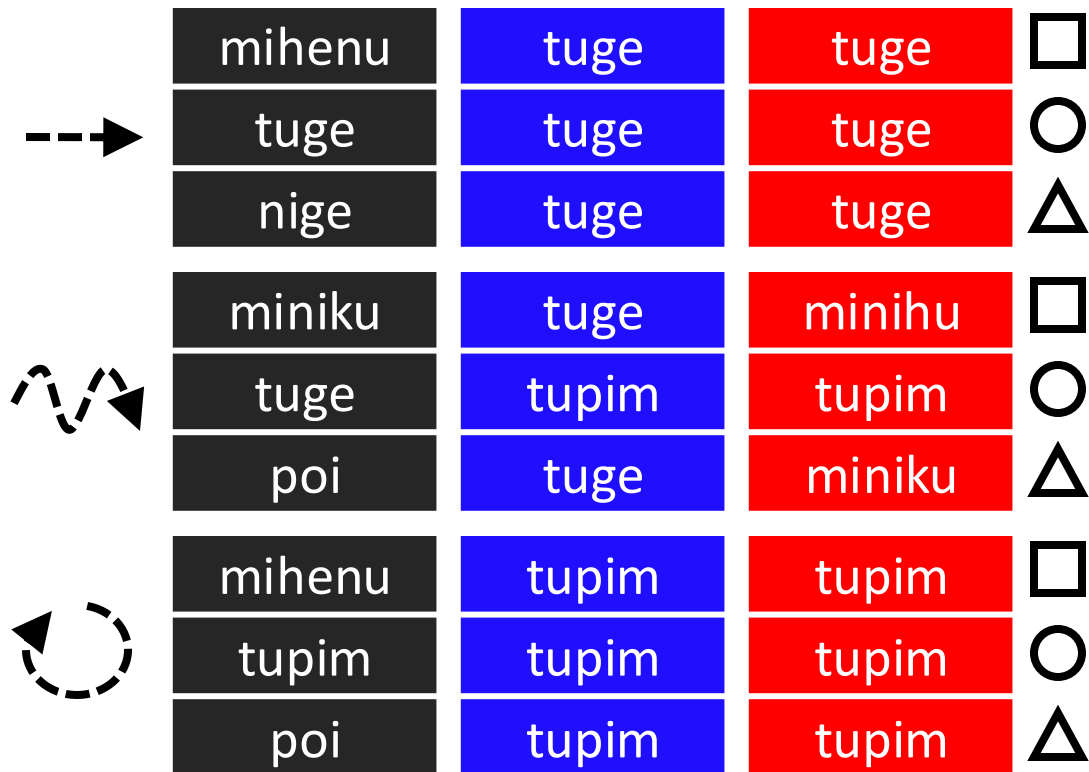
Generation 1 language (chain 4)

--→	nige	miniku	poh	□
	mip	mpo	miniku	○
	tuge	tuge	weg	△
↗↘↗	pemini	kupini	pon	□
	kimei	miwn	miheniw	○
	poi	mhip	kuwpi	△
↻	hepinimi	himini	hipe	□
	kuhepi	wige	mie	○
	pobo	tupim	hipe	△

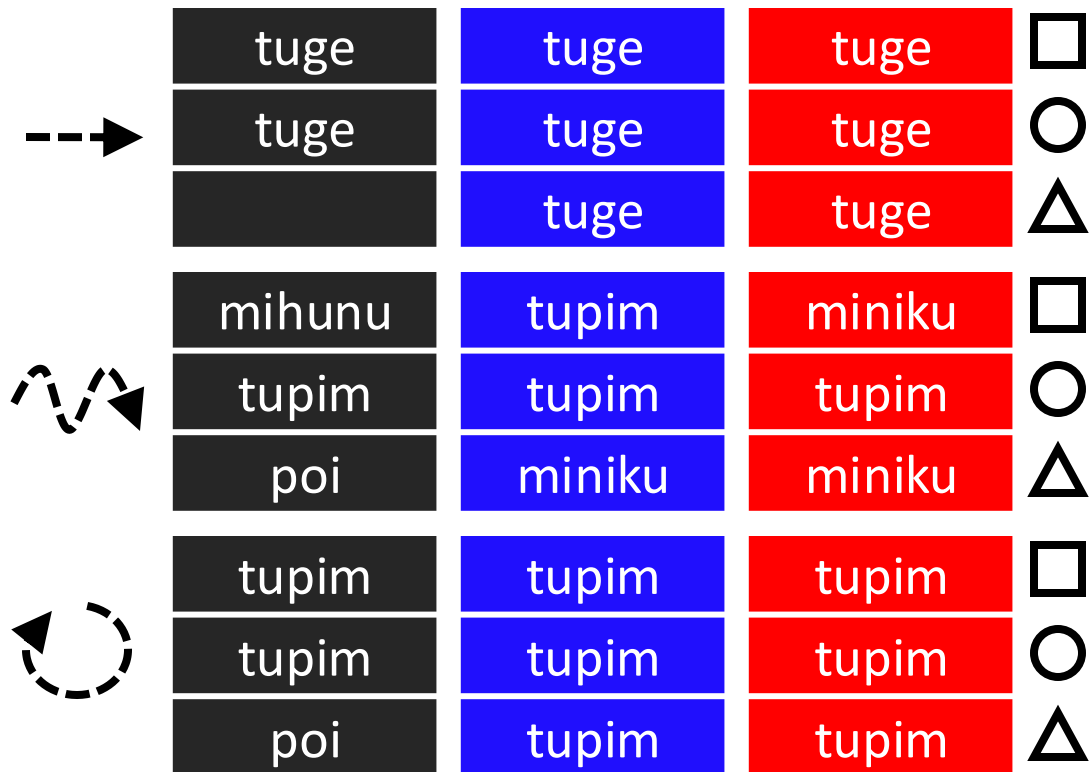
Generation 2 language (chain 4)



Generation 3 language (chain 4)



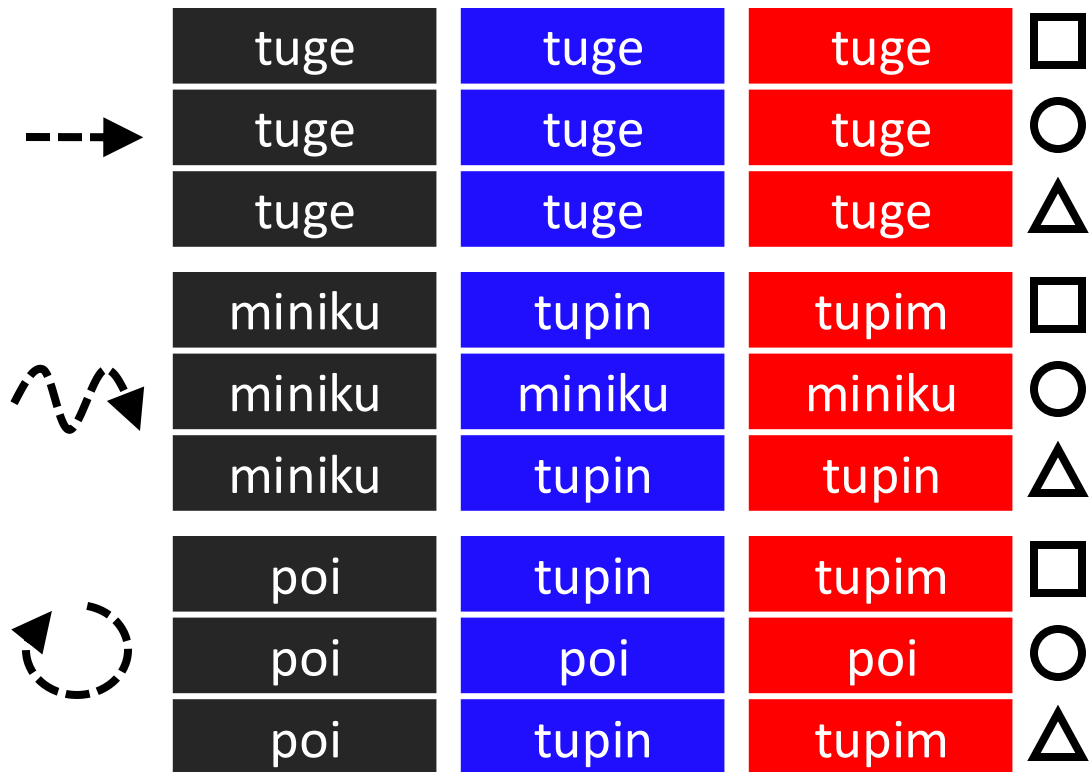
Generation 4 language (chain 4)



Generation 5 language (chain 4)

-->	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↗↘	minuhu	tupim	tupim	□
	tupim	tupim	tupim	○
	miniku	tupim	miniku	△
↻	tupim	tupim	tupim	□
	tupim	tupim	tupim	○
	poi	tupim	tupim	△

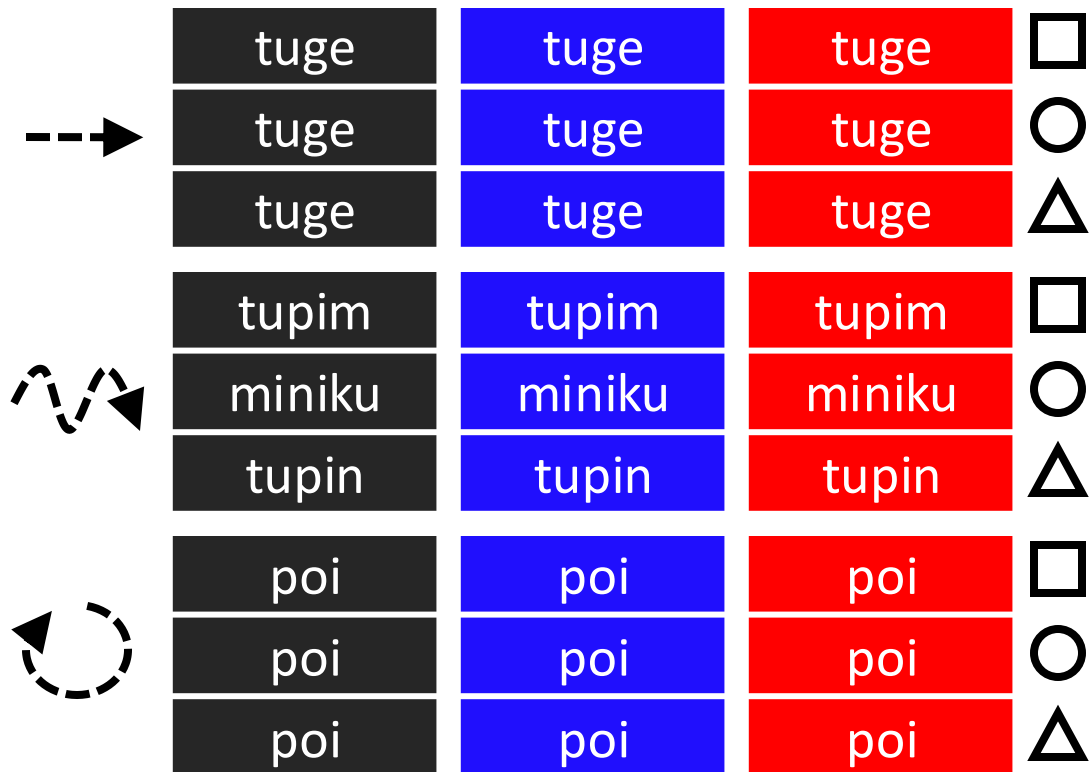
Generation 6 language (chain 4)



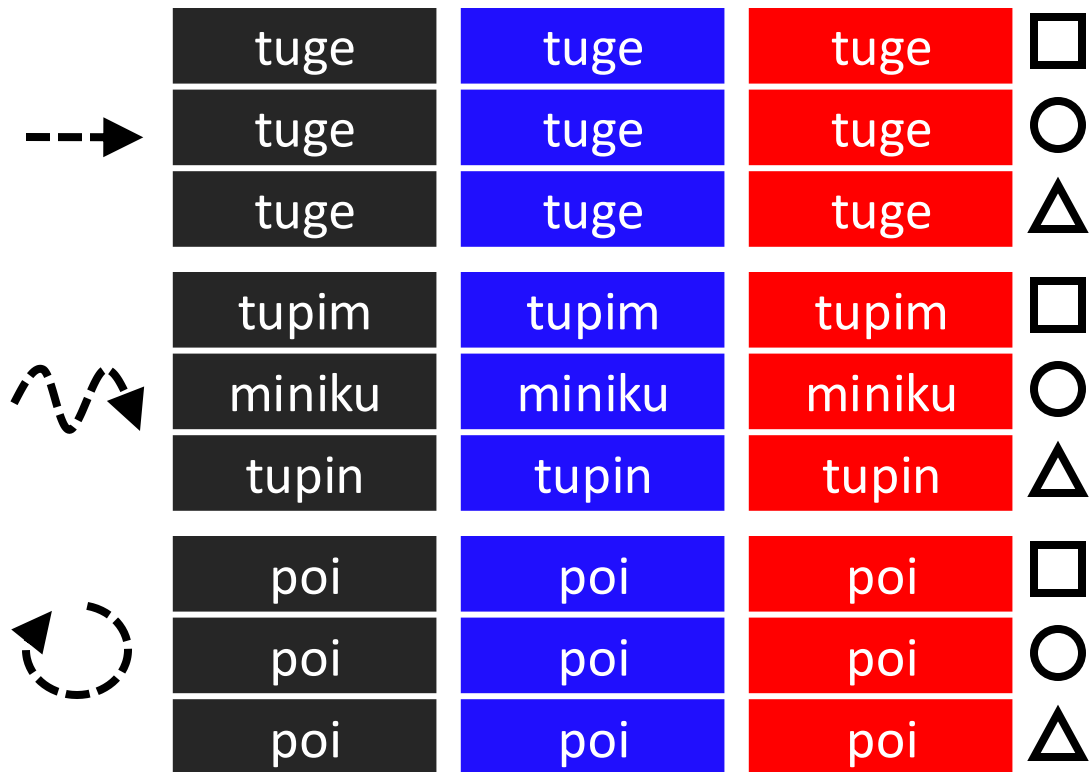
Generation 7 language (chain 4)



Generation 8 language (chain 4)



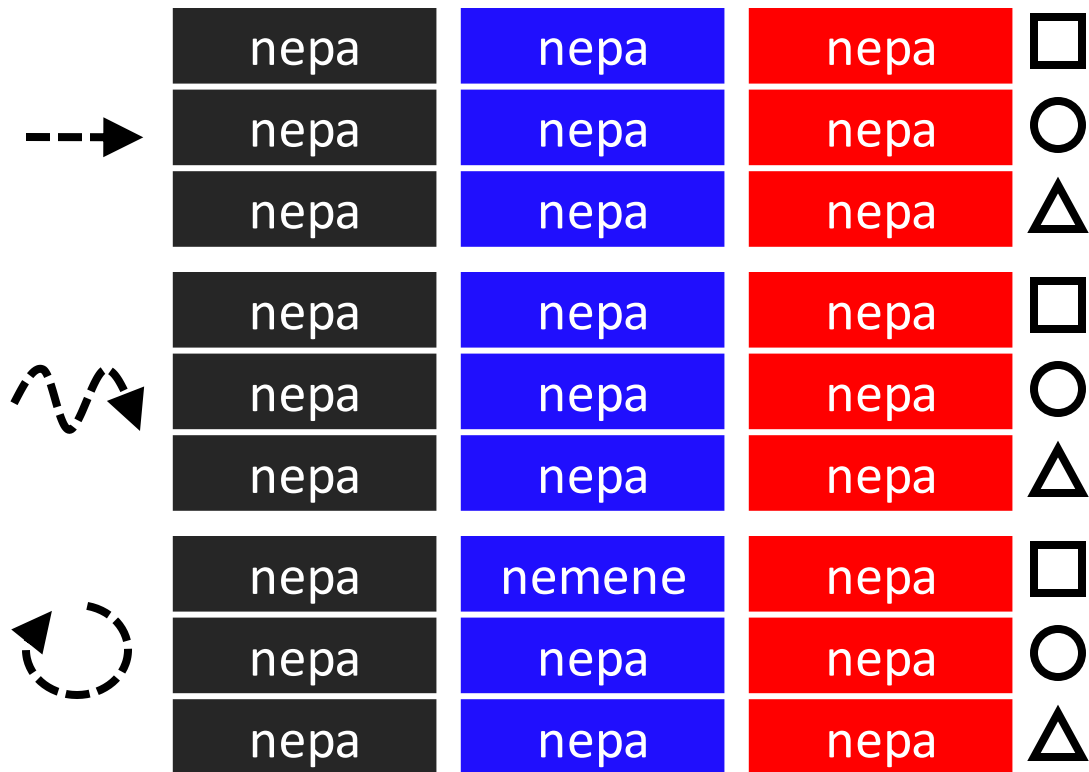
Generation 9 language (chain 4)



Generation 10 language (chain 4)

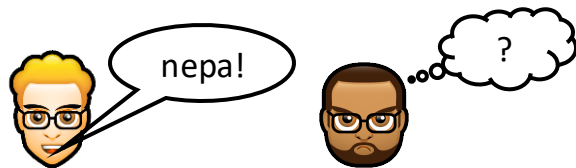
--→	tuge	tuge	tuge	□
	tuge	tuge	tuge	○
	tuge	tuge	tuge	△
↻	tupim	tupim	tupim	□
	miniku	miniku	miniku	○
	tupin	tupin	tupin	△
↻	poi	poi	poi	□
	poi	poi	poi	○
	poi	poi	poi	△

Generation 10 language (chain 1)



What happened?

The languages become **degenerate**



Generation 9 language (chain 5)

With
homonymy
my filter



ne-re-ki	la-re-ki	renana	□
ne-he-ki	la-ho-ki	re-ne-ki	○
ne-ke-ki	la-ke-ki	ra-he-ki	△



ne-re-plo	la-ne-plo	re--plo	□
ne-ho-plo	la-ho-plo	re-ho-plo	○
ne-ki-plo	la-ki-plo	ra-ho-plo	△



ne--pilu	la-ne-pilu	re--pilu	□
ne-ho-pilu	la-ho-pilu	re-he-pilu	○
ne-ki-pilu	la-ki-pilu	ra-ho-pilu	△

Beckner et al (2017)

Reanalysis/gentle roasting of Kirby, Cornish & Smith (2008)

- Their sample size was tiny
- Their statistics were very basic
- They find an interesting (?) difference between semantic dimensions

Replication

- Participants recruited from MTurk
- N=240 (2 conditions, 12 chains per condition, 10 participants per chain)
- 22-25 minutes, paid \$3

Measuring structure

“the dog **chew**-ed the bone” – “the dog **lick**-ed the bone”

Meaning distance = 1 (predicate) Signal distance = 1 (verb stem)

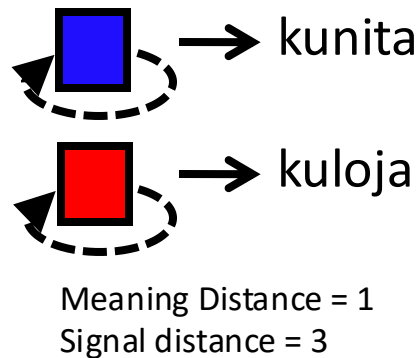
“the dog **chew**-**ed** the bone” – “the dog **lick**-**s** the bone”

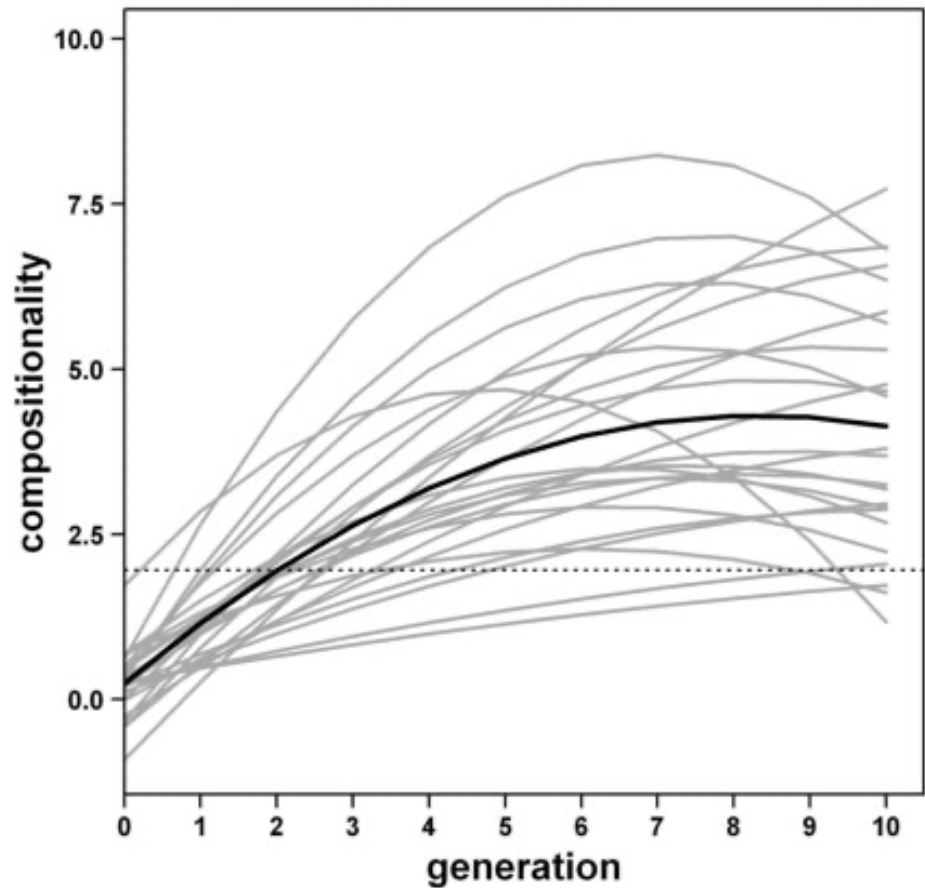
Meaning distance = 1 (predicate, tense) Signal distance = 1 (verb stem, suffix)

Pairwise meaning and signal distances will be highly correlated in a compositional system: similar meanings map to similar signals (and dissimilar meanings map to dissimilar signals)

Measuring structure

- For every pair of meaning-signal pairs
 1. Measure meaning distance (Hamming distance)
 2. Measure signal distance (Levenshtein string-edit distance)
 3. Correlate these distances
- Evaluate statistical significance of correlation
 1. Randomise label assignments, recalculate measure, repeat 1000 times to give distribution
 2. Calculate z-score of veridical correlation





	'red'	'green'	'blue'	
'berry'	shen-to	shen-ta	shen-to	'1'
	shen-tra	shen-tro	shen-tra	'2'
	shen-trio	shen-trio	shen-trio	'3'
'key'	div-tro	div-tro	div-tro	'1'
	dev-tro	dev-tro	dev-etrio	'2'
	dev-stra	div-stra	dev-stra	'3'
'phone'	lolni-tro	lolni-tro	lolni-to	'1'
	lolne-stra	lolni-tro	lolne-stro	'2'
	lolni-tra	lolni-stra	lolni-stra	'3'

Beckner et al's conclusion

Iterated learning **does** produce structure

- The original 2008 result replicates with a proper sample size
- The method also works online... but this is a challenging task on Mturk!
- Some questions:
 1. In the original paper, Kirby et al. say that cultural transmission (of language) acts as an *invisible hand*...what do they mean?
 2. There is a *bottleneck* imposed on transmission...what does that mean?

Time for Q&A/discussion on this week's reading

Next up

Lab

- Iterated learning, manipulating CSVs and looping trials

Next week

- Real-time interaction, Zipf's Law of Abbreviation